MODULE 09 · DOMAIN 1 - ARCHITECTURE BUILT FOR THE CLOUD

Hybrid DNS and Hybrid Compute

Names that resolve in both directions, and compute that runs where AWS Regions cannot.

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Hybrid DNS has **two directions**. Name the one in the question before choosing an endpoint.

Inbound or outbound endpoint?

INBOUND ENDPOINT

On premises asks AWS

- IP addresses in your VPC to query
- Resolves private hosted zone names
- Data centre DNS forwards to it
- Arrives over Direct Connect or VPN

OUTBOUND ENDPOINT

AWS asks on premises

- Lets the VPC Resolver forward out
- Used with a forwarding rule
- The rule names the domain
- Leaves over Direct Connect or VPN

THE TIEBREAKER

Who needs the answer decides the endpoint. On premises asking means inbound; AWS asking means outbound.

Rules, zones and sharing at scale

- **Forwarding rules** are shareable with RAM, but must be associated with each VPC.
- **Route 53 Profiles** bundle zones, rules and DNS Firewall, applied across VPCs and accounts.
- **Private hosted zones** answer only for associated VPCs. Associations can cross accounts.
- **Split-horizon:** public and private zones for the same name answer differently by source.
- **DNS hostnames and resolution** must be enabled on the VPC, or private zones are ignored.

> ON THE EXAM
>
> Sharing is not associating. Profiles exist because associating rules with hundreds of VPCs does not scale.

A rename, and a new resolver

- **Route 53 Resolver is now Route 53 VPC Resolver.** Same service, new name.
- It was renamed when **Route 53 Global Resolver** was introduced.
- **Global Resolver,** GA March 2026: internet-reachable, anycast, for branch and remote clients.
- It resolves **private hosted zone names** and filters malicious domains for those clients.
- **Inbound and outbound endpoints** still belong to the VPC Resolver.

> ON THE EXAM
>
> Questions may say VPC Resolver or Resolver. They are the same service. Global Resolver is a different one.

Inbound endpoint or Global Resolver?

VPC RESOLVER INBOUND ENDPOINT

A data centre on the network

- Network interfaces in your VPC
- Reached over Direct Connect or VPN
- Data centre DNS forwards to it
- Few large sites

ROUTE 53 GLOBAL RESOLVER

Many sites, one service

- Anycast, nearest Region answers
- Internet-reachable for authorised clients
- Central filtering of malicious domains
- Replaces per-site DNS appliances

THE TIEBREAKER

Hundreds of branches, central DNS filtering, no appliance per site: Global Resolver.

Hybrid compute is for workloads that must stay **near something**: equipment, users, data or a regulator.

Hybrid compute options

OPTION	WHAT IT IS	CHOOSE IT WHEN
Outposts racks	AWS hardware in your facility	Must run on premises, AWS APIs
Outposts servers	1U or 2U AWS servers	Branch, retail and edge sites
Local Zones	AWS infrastructure in a metro area	Single-digit ms to city users
EKS Hybrid Nodes	Your servers join an EKS cluster	Reliable link; AWS runs control plane
EKS Anywhere	Whole Kubernetes cluster on your hardware	Disconnected or intermittent sites
ECS Anywhere	Your servers run ECS tasks	ECS on existing hardware

ON THE EXAM

EKS Hybrid Nodes is the newest row. It needs a reliable connection; EKS Anywhere does not.

EKS Hybrid Nodes or EKS Anywhere?

EKS HYBRID NODES

Control plane in AWS

- Your servers are nodes of an EKS cluster
- AWS runs and scales the control plane
- Needs a reliable link to AWS
- Billed per vCPU hour while attached

EKS ANYWHERE

Everything on site

- Control plane and nodes run locally
- Keeps working when disconnected
- You operate the whole cluster
- Suits air-gapped or intermittent sites

THE TIEBREAKER

Disconnected, disrupted or intermittent links mean EKS Anywhere. AWS says so in the Hybrid Nodes docs.

Outposts or Local Zones?

CHOOSE OUTPOSTS WHEN

It must be on your premises

- Beside equipment or local systems
- Data must stay in your facility
- You provide space, power and cooling
- Service link of at least 500 Mbps

CHOOSE LOCAL ZONES WHEN

It must be near users

- Single-digit ms to a metro area
- AWS runs the facility
- Nothing to install or host
- Pay as in a Region

THE TIEBREAKER

On premises is a requirement for Outposts. Low latency to a city alone is a Local Zone.

One operating model across both estates

- **Systems Manager hybrid activations** manage on-premises servers like EC2.
- **ECS Anywhere** uses the SSM agent to register external instances.
- **IAM Roles Anywhere** gives on-premises workloads temporary AWS credentials.
- **CloudWatch and Managed Prometheus** give one view of both estates.
- **Long-lived access keys on premises** are the anti-pattern these replace.

> ON THE EXAM
>
> Hybrid compute is easy to start and hard to run. The answer that unifies operations usually wins.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
On premises must resolve AWS private names	Inbound endpoint
AWS must resolve on-premises names	Outbound endpoint and forwarding rule
Same DNS settings across many VPCs	Route 53 Profiles
Hundreds of branches, central DNS filtering	Route 53 Global Resolver
Site loses its link for days	EKS Anywhere, not Hybrid Nodes
Low latency to one city, nothing on premises	Local Zones

ON THE EXAM

DNS questions turn on direction and scale. Compute questions turn on connectivity and location.

> Meridian Freight's data centre applications must call internal APIs whose names exist only in a Route 53 private hosted zone. The data centre is connected to AWS by Direct Connect and runs its own DNS servers.
>
> AWS workloads do not need to resolve any on-premises names.

What should the solutions architect configure?

A. A Resolver outbound endpoint with a forwarding rule for the on-premises domain

B. A Resolver inbound endpoint, with the data centre DNS forwarding the zone's domain to it

C. Convert the private hosted zone to a public hosted zone

D. A DNS server on EC2 that copies the zone's records

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A **Outbound endpoint**

Serves the other direction: AWS asking on premises. Not needed here.

B **Inbound endpoint with conditional forwarding**

Gives on-premises DNS a target inside the VPC that answers from the zone.

C **Make the zone public**

Publishes internal names to the internet.

D **DNS on EC2**

Rebuilds the inbound endpoint by hand and drifts from the zone.

THE SIGNAL

The data centre is asking. Asking into AWS means inbound, with a conditional forwarder on premises.

> Meridian Freight has 80 VPCs across 30 accounts. Each needs the same three forwarding rules, two private hosted zones and a DNS Firewall rule group. New VPCs are created every week.
>
> Today a script shares and associates each item with each VPC, and frequently misses some.

What should the solutions architect implement?

A. Share the rules with AWS RAM, and keep the script for associations

B. A central DNS server on EC2 that every VPC uses through DHCP options

C. A Route 53 Profile containing the zones, rules and firewall group, shared and associated with the VPCs

D. Route 53 Global Resolver for all VPCs

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A RAM plus the script

Sharing was never the problem. The per-VPC associations are, and they stay scripted.

B DNS server on EC2

A self-managed single point of failure, and it bypasses DNS Firewall.

C Route 53 Profile

One bundle of zones, rules and firewall, applied as a unit to each VPC.

D Global Resolver

For branch and remote clients outside AWS, not for VPCs.

THE SIGNAL

The same DNS configuration across many VPCs and accounts is exactly what Route 53 Profiles bundle.

> Meridian Freight's 300 depots each run a small DNS appliance so that handheld scanners can resolve internal API names held in a Route 53 private hosted zone. Security wants malicious domains blocked at every depot.
>
> The company wants to retire the depot appliances and manage DNS centrally.

Which solution should the solutions architect recommend?

A. Route 53 Global Resolver with access to the private hosted zone and domain filtering

B. A VPC Resolver inbound endpoint, with a VPN from every depot

C. Make the private hosted zone public and keep filtering on the appliances

D. A VPC Resolver outbound endpoint with forwarding rules to each depot

◆ PAUSE AND DECIDE

Which solution should the solutions architect recommend?

A Global Resolver

One anycast service for every depot, resolving private names and filtering threats.

B Inbound endpoint, 300 VPNs

Works, but swaps 300 appliances for 300 VPN tunnels, and adds no filtering.

C Public zone

Exposes internal names, and keeps the appliances it was meant to retire.

D Outbound endpoint

Wrong direction: lets AWS query the depots, not the reverse.

THE SIGNAL

Hundreds of sites, central filtering, no appliance per site. Since March 2026 that is Global Resolver.

> Meridian Freight runs container workloads at remote port terminals. The terminals' satellite links drop for hours, sometimes days, and workloads must keep running and be managed locally during an outage.
>
> The company wants to standardise on Kubernetes tooling used with Amazon EKS.

Which option should the solutions architect choose for the terminals?

A Amazon EKS Hybrid Nodes

B Amazon ECS Anywhere

C An Amazon EKS cluster in the nearest AWS Local Zone

D Amazon EKS Anywhere on hardware at each terminal

◆ PAUSE AND DECIDE

Which option should the solutions architect choose for the terminals?

A **EKS Hybrid Nodes**

Control plane is in AWS, so management stops when the link drops. AWS says it does not fit these sites.

B **ECS Anywhere**

Not Kubernetes, and also relies on AWS for its control plane.

C **Local Zone**

Still reached over the satellite link, so an outage cuts the terminal off.

D **EKS Anywhere**

Whole cluster on site; keeps running and is managed locally when disconnected.

THE SIGNAL

Links that drop for days rule out every option whose control plane lives in AWS.

> Meridian Freight runs a self-managed Kubernetes cluster in its data centre, on GPU servers bought last year that must stay in service. The data centre has redundant Direct Connect links to AWS.
>
> The platform team wants to stop operating the Kubernetes control plane and use EKS tooling, with the LEAST operational overhead.

What should the solutions architect recommend?

A. Replace the cluster with Amazon EKS Anywhere on the same servers

B. Move every workload to Amazon EKS in the Region and retire the GPU servers

C. Amazon EKS Hybrid Nodes, joining the GPU servers to an EKS cluster

D. AWS Outposts racks to replace the GPU servers

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A EKS Anywhere

The control plane still runs on site, so the team keeps operating it.

B Move everything to the Region

Retires servers the scenario says must stay in service.

C EKS Hybrid Nodes

AWS runs the control plane; the existing GPU servers become nodes.

D Outposts

Replaces hardware that must be kept, at significant new cost.

THE SIGNAL

A reliable link, hardware that must stay, and no control plane to run. That is the Hybrid Nodes case.

> Meridian Freight is launching a live dispatch app for drivers in one large city, several hundred kilometres from the nearest AWS Region. The app needs single-digit millisecond latency to drivers' devices.
>
> Meridian has no data centre in that city, and no data needs to stay on premises.

Where should the latency-sensitive tier run?

A An AWS Local Zone in that city

B An AWS Outposts rack in a rented colocation facility in the city

C The nearest Region, behind Amazon CloudFront

D The nearest Region, behind AWS Global Accelerator

◆ PAUSE AND DECIDE

Where should the latency-sensitive tier run?

A Local Zone

AWS compute in the city itself, with nothing to host or maintain.

B Outposts in a colocation facility

Achieves the latency, but adds hardware, space and a service link for no requirement.

C Region with CloudFront

Caches content, but dispatch is dynamic; each request still travels to the Region.

D Region with Global Accelerator

Improves the path, but cannot remove hundreds of kilometres of distance.

THE SIGNAL

Low latency to a city with nothing that must be on premises is a Local Zone. Outposts needs a reason to be on site.

What to carry out of this module

- **Inbound: on premises asks AWS. Outbound: AWS asks on premises.**
- **Route 53 Profiles** for DNS at scale. **Global Resolver** for many remote sites.
- **Route 53 Resolver is now called VPC Resolver.** Same service.
- **Reliable link:** EKS Hybrid Nodes. **Intermittent link:** EKS Anywhere.
- **Must be on premises:** Outposts. **Must be near a city:** Local Zones.

> ON THE EXAM
>
> Two changes since 2025: the VPC Resolver rename, and Global Resolver going GA in March 2026.